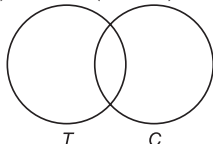


$$= 50 - 12 = 38.$$

$$n(B) = 25, n(H) = 16, n(B \cup H) = 38.$$

$$\begin{aligned} \therefore \text{Number of students who take both Bengali and Hindi} \\ &= n(B \cap H) = n(B) + n(H) - n(B \cup H) \\ &= (25 + 16) - 38 = (41 - 38) = 3. \end{aligned}$$

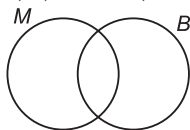
$$454. n(T) = 60, n(C) = 40, n(T \cap C) = 25.$$



$$\begin{aligned} \therefore \text{Number of students who take either tea or coffee or both} \\ &= n(T \cup C) = n(T) + n(C) - n(T \cap C) \\ &= 60 + 40 - 25 = 100 - 25 = 75. \end{aligned}$$

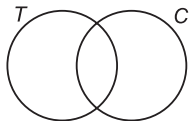
$$\therefore \text{Required number of students} = (100 - 75) = 25.$$

$$455. n(M \cup B) = 25, n(M) = 12, n(M - B) = 8.$$



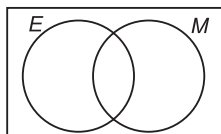
$$\begin{aligned} \text{Number students who have taken both Mathematics and} \\ \text{Biology} &= n(M \cap B) = n(M) - n(M - B) = 12 - 8 = 4. \end{aligned}$$

$$456. n(T \cup C) = 52, n(T - C) = 16, n(T) = 33.$$



$$\begin{aligned} \text{Number of persons who drink coffee but not tea} \\ &= n(C - T) = n(T \cup C) - n(T) = 52 - 33 = 19. \end{aligned}$$

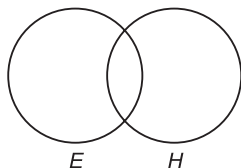
$$457. n(E) = 3000, n(M) = 2000, n(E \cap M) = 1500.$$



$$\begin{aligned} n(E \cup M) &= n(E) + n(M) - n(E \cap M) = 3000 + 2000 - 1500 \\ &= 3500. \end{aligned}$$

$$\begin{aligned} \text{Number of pure vegetarians} &= 4000 - n(E \cup M) \\ &= 4000 - 3500 = 500. \end{aligned}$$

$$458. n(E \cup H) = 50; n(E \cap H) = 10; n(E) = 21.$$



$$\begin{aligned} \text{Number of students who can speak Hindi} &= n(H) \\ &= n(E \cap H) + n(H - E) = 10 + 39 = 49. \end{aligned}$$

$$\begin{aligned} \text{Number of students who can speak only Hindi} \\ &= n(H - E) = n(H) - n(E \cap H) = 49 - 10 = 39. \end{aligned}$$

$$\begin{aligned} \text{Number of students who can speak only English} \\ &= n(E - H) = n(E) - n(E \cap H) = 21 - 10 = 11. \end{aligned}$$

$$459. \text{Let the total number of staff members be } x.$$

Then, the number who can type or take shorthand

$$= \left(x - \frac{3x}{4} \right) = \frac{x}{4}.$$

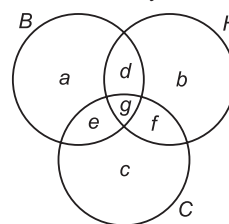
Let A and B represent the sets of persons who can type and take shorthand respectively.

$$\text{Then, } n(A \cup B) = \frac{x}{4}, n(A) = \frac{x}{5} \text{ and } n(B) = \frac{x}{3}.$$

$$n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

$$= \left(\frac{x}{5} + \frac{x}{3} - \frac{x}{4} \right) = \left(\frac{12x + 20x - 15x}{60} \right) = \frac{17x}{60}.$$

460. Let circles B , H and C represent the number of players who play basketball, hockey and cricket respectively.



$$\text{Then, } a + d + e + g = 20 \quad \dots(i)$$

$$b + d + f + g = 24 \quad \dots(ii)$$

$$c + e + f + g = 27 \quad \dots(iii)$$

$$d + g = 12 \quad \dots(iv)$$

$$e + g = 10 \quad \dots(v)$$

$$f + g = 14 \quad \dots(vi)$$

$$g = 7 \quad \dots(vii)$$

From (iv), (v), (vi) and (vii), we have:

$$d = 12 - g = 12 - 7 = 5;$$

$$e = 10 - g = 10 - 7 = 3;$$

$$f = 14 - g = 14 - 7 = 7.$$

From (iii), we have: $c = 27 - (e + f + g)$

$$= 27 - (3 + 7 + 7) = 10.$$

From (ii), we have: $b = 24 - (d + f + g)$

$$= 24 - (5 + 7 + 7) = 5.$$

From (i), we have: $a = 20 - (d + e + g)$

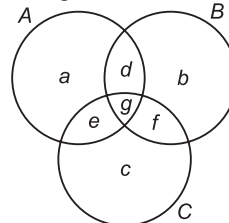
$$= 20 - (5 + 3 + 7) = 5.$$

Hence, total number of players = $a + b + c + d + e + f + g$

$$= (5 + 5 + 10 + 5 + 3 + 7 + 7)$$

$$= 42.$$

461. Let the circles A , B and C represent the aliens having two noses, three legs and four ears respectively.



$$\text{Then, } a + d + e + g = 40 \quad \dots(i)$$

$$b + d + f + g = 30 \quad \dots(ii)$$

$$c + e + f + g = 20 \quad \dots(iii)$$

$$d + g = 10 \quad \dots(iv)$$

$$f + g = 12 \quad \dots(v)$$

$$e + g = 5 \quad \dots(vi)$$

$$g = 3 \quad \dots(vii)$$

From (iv), (v), (vi) and (vii), we have:

$$d = 10 - g = 10 - 3 = 7;$$

$$e = 5 - g = 5 - 3 = 2;$$

$$f = 12 - g = 12 - 3 = 9.$$